



Psychopharmaceutical treatment for neurobehavioral problems in Duchenne muscular dystrophy: a descriptive study using real-world data

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ABSTRACT

Patients with Duchenne muscular dystrophy (DMD) are at risk to develop neurobehavioral problems. Evidence on how to treat these difficulties is scarce. This descriptive study reports the clinical experience with psychopharmaceutical treatment in 52 patients with DMD.

Electronic patient files were searched for patients with DMD that had been treated with psychopharmaceuticals between 2008 and 2022. Information about neurobehavioral symptoms, type of medication, side effects, and behavioral changes were collected. Two independent clinicians used the clinical global impression scale (CGI) to assess severity of the neurobehavioral problems before and the change in symptoms after treatment. Descriptive statistics were used.

Our results include 52 males with DMD (mean age 11 years) treated with psychopharmaceuticals of which 55.8% had four or more comorbid neurobehavioral symptoms. The clinical condition was much improved on the CGI in 54.2% treated with methylphenidate, in 38.9% of the patients treated with fluoxetine, and in 22.2% treated with risperidone. Minimal effects and side effects were also reported.

In conclusion, patients with DMD may experience severe neurobehavioral symptoms interfering with learning and/or development. Treatment with psychopharmaceuticals can improve these neurobehavioral symptoms, but further research is needed to gain better insights in psychopharmaceutical treatment in patients with DMD.

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1. Introduction

Duchenne muscular dystrophy (DMD) is a progressive neuromuscular disorder caused by a mutation in the dystrophin gene on the X-chromosome leading to deficient expression of the protein dystrophin [1]. In skeletal muscle cells, dystrophin deficiency results in progressive muscle damage, muscle atrophy and eventually a decline of motor function. Dystrophin isoforms, however, are not only expressed in skeletal muscles, but also in

the brain [2]. Patients with DMD have a higher risk for intellectual disability, learning disorders and psychiatric disorders [3–7]. Both cognitive and behavioral difficulties seem to be related with the specific genetic defect. Patients with a more distal defect in the dystrophin gene, hence affecting the production of brain dystrophin isoform Dp140 and Dp71, seem to have a higher risk for intellectual problems and psychiatric difficulties [8,9]. Still, the exact relationship between dystrophin expression, brain structure and brain functioning, and behavioral and cognitive outcomes remain unclear.

The combination of physical decline and brain-related comorbidities can lead to complex behavioral and cognitive phenotypes (referred to as neurobehavioral in this paper) with difficulties in family, social and school context. Multiple studies

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have described these difficulties, but scientific evidence about how to manage these problems or which psychological interventions are recommended, is scarce. In the international guidelines about the diagnosis and management of DMD is stated that patients with DMD should be regularly screened for neurobehavioral problems, but what to do when these are detected, is not specified [10]. The guidelines, however, suggest that psychopharmaceutical drugs (PDs) could be of help in those cases where the behavioral or cognitive phenotype is problematic on a level that it disturbs the child's development and the daily functioning of the child and the surroundings [10]. Anecdotal evidence of PD use in DMD patients suggests that PD can be safe and efficient in some cases [11,12]. One large study ($N=700$) supports these findings, but details on side effects, improvements, and efficacy are missing [13].

Overall, the scarcity of data on psychopharmaceutical interventions in neuromuscular disorders is in contrast to the frequency of reported psychiatric and neurodevelopmental comorbidities and the increase in PD use over the last decades across multiple pediatric populations [14–18]. More research on clinical experience and clinical efficacy of interventions with PD within the DMD population is needed to further improve the care for these patients.

The primary aim of this study is to describe clinical experience and side effects of treatment with PD in adolescents with DMD using real-world data.

2. Methodology

2.1. Study design

This multicenter retrospective clinical series study has a descriptive design, analysing data extracted from the clinics' electronic health records. Data from two different centers were obtained by a retrospective search through the patient's records. The center for Neurological Learning Disabilities (CNL) at Kempenhaeghe (The Netherlands) is an outpatient clinic specialized in diagnostics and treatment for neurocognitive and neuropsychiatric comorbidities e.g. for neuromuscular disorders such as DMD. As part of regular care patients received an extensive medical (performed by treating neurologist and psychiatrist (SK and PC) and cardiologist) and neuropsychological assessment (performed by JH). Neuropsychiatric comorbidities were established based on (1) a psychiatric interview based on the Diagnostic and Statistical Manual of Mental Disorders-V (DSM-V) criteria, (2) parent and teacher reports measuring emotional, behavioral and cognitive symptoms, and (3) an extensive cognitive assessment (using various evidence-based neuropsychological tests assessing cognitive impairment, including those measuring language, learning, and memory).

The Neuromuscular Reference Center (NMRC) for Children from the University Hospitals Leuven (Belgium), provides multidisciplinary care for children with neuromuscular disorders and their families. Boys with DMD are assessed by both a child neurologist with expertise in neuromuscular disorders in children (LDW) and a neuropsychologist (SG) on a regular basis in order to screen for neurobehavioral difficulties and start early intervention. In case of positive screening, the patients are formally assessed within the NMRC or referred to a specialized regional center. Assessment procedure is indicated by the symptoms and case-specific, but always starts with a exhaustive diagnostic interview by a child neurologist and/or a child neuropsychologist, which, if indicated, can be followed by formal neuropsychological testing (IQ, memory, attention span, executive functioning), self- and proxy-reported questionnaires, a psychiatric consult and/or school observation. All assessment results and interventions are recorded in the electronic patient files.

This study was conducted according to the advice of the medical ethical committees of both centers.

2.2. Participants

The inclusion criteria for this study were 1) male patients, 2) with a genetic confirmation of DMD, 3) and available data on PD use at baseline (T0) and follow-up (T1) between 2008 and 2022. Glucocorticoid (GC) therapy is standard of care in both Belgium and the Netherlands, which is in accordance with the international guidelines.

2.3. Data collection

Patient records who met inclusion criteria were filtered. The following parameters were collected and inserted into a database: age at which the neurobehavioral symptoms started, nature of the neurobehavioral symptoms, age at initiation of pharmaceutical intervention, psychopharmaceutical product used, medication regimen, medication dose at start and end of the intervention, age at interruption of the treatment (if applicable), reason for interruption (if applicable), changes in regimen or dose (if applicable), type of steroid used (if applicable), and mutation site. Based on the mutation site three groups were formed: (1) any mutation upstream of intron 44 (Dp140 present), (2) any mutation between exon 45 and 50 (Dp140 expression unknown), (3) any mutation from exon 51 to 62 (Dp140 absent), and (4) any mutation downstream from exon 63 (Dp140-/Dp71-). Patients treated with two types of the same psychopharmaceutical product (e.g. short-working and long-working methylphenidate) were counted only once.

To classify neurobehavioral problems, developmental, emotional and behavioral symptoms are described according to the neurodevelopmental domains grouped by Darmahkasi [13]: autistic features, inattention, hyperactivity, learning difficulties, language delay, global intellectual delay, vocal and/or motor tics, emotional and behavioral dysregulation, obsessive compulsive features, anxiety features, and depressive symptoms, and one extra category: sleep problems.

2.4. Assessments

To evaluate the effect of the psychopharmaceutical intervention after initiation on T0, the clinical information reported at the follow-up consult T1 was analysed. When severe side effects were reported, the date of recording of these side effects was marked as T1. To quantify the therapeutic effect at T1 the Clinical Global Impression scale (CGI) was used [19]. The CGI is developed for clinicians to rate the patient's global functioning and to assess the effect of study medication in clinical practice [19]. It is widely used in psychiatry and provides a 3-item observer-rated scale of which the illness severity scale (CGI-S) and the global improvement or change scale (CGI-C) were used in this study. The CGI-S uses the question "Considering your total clinical experience with this particular population, how mentally ill is the patient at this time?". In this study, this question was answered based on the neurobehavioral difficulties described at T0. Examiners had to independently rate the situation on a scale from 0 to 7 (0 = not assessed; 1 = normal, not at all ill; 2 = borderline mentally ill; 3 = mildly ill; 4 = moderately ill; 5 = markedly ill; 6 = severely ill; 7 = among the most extremely ill patients).

The CGI-C rates the condition on T1 after the start of the pharmaceutical intervention. Problems are rated, similarly to the CGI-S, on a scale from 0 to 7 (0 = not assessed; 1 = very much improved; 2 = much improved; 3 = minimally improved; 4 = no change; 5 = minimally worse; 6 = much worse; 7 = very

much worse). When at T1 the condition of a patient was rated as minimally, much or very much worse, the first two authors (PW and SG) analyzed the electronic patient files to record (side-)effects of the PD prescribed.

Cohen's kappa was calculated to determine the level of agreement between the two independent reviewers (P.C. and S.K., and S.G. and L.D.W.). If initially no consensus was reached, the two raters discussed the case and came to an agreement. Doubtful cases were examined by a third rater (J.H.). When there was a one-point difference between the ratings an average score was used.

2.5. Statistical analyses

Descriptive statistics were presented as the mean, standard deviation and percentages. Two MANOVA's with sum of behavioral symptoms and score on the CGI-S as dependent variables and respectively corticosteroid regime and mutation group as fixed factors were calculated to explore if corticosteroid treatment or mutation site was related to behavioral functioning in our sample. All analyses were carried out using SPSS (SPSS, Inc.) version 26 and Microsoft Excel (version 16.64).

3. Results

3.1. Patient characteristics

Reported patients were all males with DMD that used PD to alleviate comorbid neurodevelopmental symptoms. In total 52 patients could be included (mean age at start intervention=11.00, $SD=4.6$, range=5–23 years). Mean age on which problems started was 9.42 years ($SD=4.9$, range=2–22). None of the patients had a mutation downstream from exon 63. Patients' demographics are summarized in Table 1.

3.2. CGI-S and CGI-C assessment

A minimal level of agreement was found between the two reviewers on both CGI-S and CGI-C respectively, $k=0.3$ and $k=0.2$. If initially no consensus was reached, the two raters discussed the case and came to an agreement. We agreed that when there was a one-point difference between the ratings an average score was

used. It should be noted that one of our raters consistently marked patients less severely ill than the other rater.

3.3. Neurobehavioral symptoms

Table 2 shows the frequency of the different neurobehavioral symptoms treated in this population. Inattentive features were treated the most (63.5%) and depressive symptoms the least frequent (9.6%) in the total sample. In patients treated with prednisolone, inattentive features, hyperactivity and sleep problems were the most reported behavioral symptoms, whereas, for patients treated with deflazacort, the most reported behavioral symptoms were emotional and behavioral dysregulation, obsessive and compulsive features, and anxiety features. Most patients had more than one neurobehavioral symptom. Table 3 demonstrates these comorbidities by displaying how many symptoms were seen in one patient. 55.8% of the patients had four or more different neurobehavioral symptoms. A MANOVA with number of symptoms and CGI-S score as dependent variable and corticosteroid treatment as independent variable (Wilks' Lambda=0.907, $F(4,78)=0.975$, $p=0.426$, $\eta^2=0.05$) showed no significant effect. Additionally, we didn't find a significant effect with a MANCOVA with number of symptoms and CGI-S score as dependent variable and mutation group as independent variable (Wilks' Lambda=0.922, $F(4,86)=0.886$, $p=0.476$, $\eta^2=0.04$).

3.4. Psychopharmaceutical drugs

In this sample of 52 DMD patients, a total of 66 unique PD administrations were identified. 43 patients only used 1 PD, six used 2 different PDs, two patients had 3 types of PDs administered, and one patient got 6 different PDs prescribed. Table 4 gives an overview of the psychopharmaceutical drugs that were used to treat neurobehavioral difficulties in our patient sample, and gives information about the indication, dose, severity of the indication and the estimated change on T1. CGI-S scores were reported per patient, CGI-C scores were given per PD treatment. So, if a patient received two types of antipsychotics both treatments were rated on the change index. In total 42.4% ($N=28$) PD-treatments were stopped of which 32.1% ($N=9$) due to side effects, 46.4% ($N=13$) no perceived effect, 21.4% ($N=6$) growing out of deficit.

3.4.1. Stimulants

Twenty-five patients were treated with stimulant drugs. Two patients received two or three types of stimulants. One patient was treated first with methylphenidate, followed by dexamphetamine and lisdexamphetamine (with one month overlap), the other with methylphenidate followed by atomoxetine.

The first neurobehavioral symptoms of these 25 patients were reported at an average age of 6.7 years ($SD=1.7$) and stimulant treatment was started more than one year later (mean age= 8.3, $SD=1.6$).

A few changes in dosing or regimen were made: two patients changed from short- to long-acting methylphenidate because of rebound effects, one patient changed from long- to short-acting methylphenidate because of weight loss, and in the case of dexamphetamine the dose was decreased because of difficulties falling asleep.

In six patients the stimulant treatment (21.4%) was stopped because of side effects e.g. stomach issues ($N=1$), weight loss ($N=3$), trouble sleeping ($N=1$), obsessive/compulsive behavior ($N=1$), adverse drug effect ($N=1$). One of these patients had both stomach issues and trouble sleeping. Eight treatments (32%) were stopped due to other reasons e.g. no perceived effect ($N=3$), growing out of deficit ($N=5$). See Table 5 for an overview of the CGI-C scores of stimulant treatments in DMD. In 54.2% of

Table 1
Patient characteristics.

Variable	N	%
Country		
The Netherlands	34	65.4
Belgium	18	34.6
Steroid use		
Yes	48	92.4
No	2	3.8
Not anymore	2	3.8
Type of steroid		
Prednisone	24	46.2
Deflazacort	24	46.2
Unknown	4	7.7
Psychopharmaceutical treatment		
Yes	30	57.7
Not anymore	22	42.3
Genotype subgroups		
Dp140+	16	30.8
Dp140?	12	23.1
Dp140-	24	46.2

Note. Dp140+ = mutation upstream of intron 44; Dp140? = mutation between exon 45 and 50; Dp140- = mutation downstream from exon 51, N = number.

Table 2
Frequency of different neurobehavioral symptoms.

Neurobehavioral symptoms	Total group (N=52)	Steroid regime Prednisone (N=24)	Deflazacort (N=24)	Unknown (N=4)
Inattentive features	33 (63.5)	21 (87.5)	10 (41.7)	2 (50.0)
Emotional and behavioral dysregulation	31 (59.6)	14 (58.3)	14 (58.3)	3 (75.0)
Sleep problems	27 (51.9)	15 (62.5)	10 (41.7)	2 (50.0)
Hyperactivity features	23 (44.2)	15 (62.5)	7 (29.2)	1 (25.0)
Obsessive and compulsive features	20 (38.5)	8 (33.3)	12 (50.0)	0 (0.0)
Learning difficulties	19 (36.5)	11 (45.8)	7 (29.2)	1 (25.0)
Anxiety features	19 (36.5)	7 (29.2)	11 (45.8)	1 (25.0)
Autistic features	18 (34.6)	9 (37.5)	7 (29.2)	2 (50.0)
Language delay	16 (30.8)	14 (58.3)	2 (8.3)	0 (0.0)
Motor and/or vocal tics	8 (15.4)	3 (12.5)	2 (8.3)	3 (75.0)
Global intellectual delay	6 (11.5)	3 (12.5)	2 (8.3)	1 (25.0)
Depressive symptoms	5 (9.6)	3 (12.5)	2 (8.3)	0 (0.0)

Note. N = number. Frequencies are given, with percentage between parentheses.

Table 3
Frequencies of neurobehavioral symptoms.

Number of symptoms	Total Group (N=52)	Steroid regime Prednisone (N=24)	Deflazacort (N=24)	Unknown (N=4)
1	2 (3.8)	0 (0.0)	2 (8.3)	0 (0.0)
2	9 (17.3)	3 (12.5)	4 (16.7)	2 (50.0)
3	12 (23.1)	4 (16.7)	8 (33.3)	0 (0.0)
4	6 (11.5)	3 (12.5)	3 (12.5)	0 (0.0)
5	6 (11.5)	2 (8.3)	3 (12.5)	1 (25.0)
6	6 (11.5)	3 (12.5)	3 (12.5)	0 (0.0)
7	9 (17.3)	7 (29.2)	1 (4.2)	1 (25.0)
8	2 (3.8)	2 (8.3)	0 (0.0)	0 (0.0)

Note. N = number. Frequencies are given, with percentage between parentheses.

Table 4
Overview of used psychotropic drugs, indications and global improvement or change in behavioral symptoms.

	N	%	Age	Number of symptoms	ASF	INA	HYP	LEA	LAN	INT	TIC	EMB	OCF	ANX	DEP	SLE	Start dose*	End dose**	CGI-S	CGI-C
Stimulants	28	42.4	8.3 (1.6)	4.7 (1.7)	7	26	19	15	9	5	4	19	5	7	0	16			4.0 (0.9)	2.6 (0.8)
Methylphenidate	24		8.0 (1.4)	4.6 (1.7)	6	22	16	15	9	5	2	15	4	5	0	12	0.41 (0.38)	14.6 (8.0)	3.9 (0.9)	2.5 (0.6)
Atomoxetine	2		11.5 (0.7)	6.0 (2.8)	1	2	1	0	0	0	1	2	1	2	0	2	0.76 (0.71)	34.5 (7.8)	4.5 (0.7)	3.5 (2.1)
Dexamphetamine	1		10.0	4	0	1	1	0	0	0	0	1	0	0	0	1	0.37	5.0	4.5	3.5
Lisdexamphetamine	1		9.0	5	0	1	1	0	0	0	1	1	0	0	0	1	0.54	20.0	4.5	3.0
Antidepressants	22	33.3	14.2 (4.2)	3.9 (2.1)	5	9	5	2	5	0	4	14	14	11	5	12			4.2 (0.6)	2.9 (1.0)
Fluoxetine	18		14.0 (4.5)	3.6 (2.0)	4	6	3	2	3	0	4	11	11	9	3	8	0.35 (0.29)	14.2 (5.6)	4.2 (0.6)	3.0 (1.0)
Citalopram	1		13.0	7.0	0	1	0	0	1	0	0	1	1	1	1	1	0.22	10.0	5.0	4.0
Amitriptyline	1		15.0	5.0	0	1	1	0	0	0	0	0	1	0	1	1	0.51	50.0	5.0	2.0
Escitalopram	1		14.0	7.0	1	1	1	0	1	0	0	1	1	0	0	1	0.51	20.0	4.0	2.0
Sertraline	1		19.0	3.0	0	0	0	0	0	0	0	1	0	1	0	1	0.47	25.0	4.0	2.0
Antipsychotics	14	21.2	10.2 (5.2)	5.1 (1.8)	10	8	8	6	6	4	1	11	4	4	1	8			4.8 (0.8)	3.0 (1.1)
Risperidone	9		12.3 (5.6)	4.8 (2.1)	6	5	3	3	3	2	1	7	3	3	1	6	0.01 (0.01)	22.0 (37.0)	4.7 (0.6)	2.9 (0.8)
Pipamperone	3		6.3 (1.5)	5.3 (1.2)	2	2	3	2	2	1	0	3	0	0	0	1	0.89 (0.46)	20.7 (17.5)	4.7 (1.3)	3.7 (1.3)
Aripiprazole	2		8.0 (2.8)	6.0 (0.0)	2	1	2	1	1	1	0	1	1	1	0	1	0.32 (0.02)	1.5 (0.7)	5.5 (0.7)	2.5 (1.1)
Tranquilizers	2	3.0	21.5 (2.1)	4.5 (3.5)	0	1	0	1	0	0	1	1	1	2	1	1			5.3 (1.1)	2.5 (0.7)
Alprazolam	1		23.0	2.0	0	0	0	0	0	0	1	0	0	1	0	0	0.30	15.0	6.0	2.0
Oxazepam	1		20.0	7.0	0	1	0	1	0	0	0	1	1	1	1	1	0.25	15.0	4.5	3.0

Note:.

* doses in mg/kg/day;

** doses in mg/day; means are given with standard deviations between parentheses, ASF autistic features, INA inattention, HYP hyperactivity, LEA learning difficulties, LAN language delay, INT global intellectual delay, TIC vocal and/or motor tics, EMB emotional and behavioral dysregulation, OCF obsessive compulsive features, ANX anxiety features, DEP depressive symptoms, SLE sleep problems, CGI-S illness severity (0 = not assessed; 1 = normal, not at all ill; 2 = borderline mentally ill; 3 = mildly ill; 4 = moderately ill; 5 = markedly ill; 6 = severely ill; 7 = among the most extremely ill patients), CGI-C global improvement or change (0 = not assessed; 1 = very much improved; 2 = much improved; 3 = minimally improved; 4 = no change; 5 = minimally worse; 6 = much worse; 7 = very much worse).

the patients treated with methylphenidate the condition was much improved (score 2 on CGI-C) compared to on admission, in 91.7% the condition was at least minimally improved (score 2 or 3 on CGI-C), and in 8.3% of the patients the treatment had no effect (score 4 on CGI-C). In one patient treated with atomoxetine the condition was minimally worse (score 5 on CGI-C), and in another patient it was much improved (score 2 on the CGI-C), while methylphenidate had no effect on him (score 4 on CGI-C). Dexamphetamine had a minimally improving/no effect

on the patient's condition since admission (score 4 on CGI-C), lisdexamphetamine had also only minimal effect on him (score 3 on CGI-C).

3.4.2. Antidepressants

Antidepressants were the second most frequently prescribed group of PDs in this DMD sample (22 treatments in 21 patients). Neurobehavioral symptoms for which these PDs were prescribed started at an average age of 11.6 years ($SD = 5.5$, range = 4–21),

Table 5
CGI-C scores of psychopharmaceuticals in DMD.

		CGI-C	%	Cum.%
Stimulants	Methylphenidate	2.0	54.2	54.2
		2.5	16.7	70.8
		3.0	20.8	91.7
		4.0	8.3	100.0
	Atomoxetine	2.0	50.0	50.0
		5.0	50.0	100.0
	Dexamphetamine	3.5	100.0	100.0
	Lisdexamphetamine	3.0	100.0	100.0
Antidepressants	Fluoxetine	2.0	33.3	33.3
		2.5	5.6	38.9
		3.0	33.3	72.2
		3.5	5.6	77.8
		4.0	16.7	94.4
		6.0	5.6	100.0
	Citalopram	4.0	100.0	100.0
	Amitriptyline	2.0	100.0	100.0
	Escitalopram	2.0	100.0	100.0
	Sertraline	2.0	100.0	100.0
Antipsychotics	Risperidone	2.0	22.2	22.2
		2.5	33.3	55.6
		3.0	11.1	66.7
		3.5	11.1	77.8
		4.0	22.2	100.0
	Pipamperone	2.5	33.3	33.3
		3.5	33.3	66.7
		5.0	33.3	100.0
	Aripiprazole	1.0	50.0	50.0
		4.0	50.0	100.0
Tranquilizers	Alprazolam	2.0	100.0	100.0
	Oxazepam	3.0	100.0	100.0

Note. CGI-C global improvement or change (0= not assessed; 1= very much improved; 2= much improved; 3= minimally improved; 4= no change; 5= minimally worse; 6= much worse; 7= very Much worse), Cum. = cumulative.

while the average age on which PD treatment was started was 14.2 years ($SD= 4.2$, range= 9–23).

Six patients stopped using fluoxetine. Four of them because an insufficient reduction in behavior or feelings of anxiety (18.2%). Two of them (9.1%) experienced side effects like anger issues ($N=1$) or compulsions with self-harming behavior and hearing an inner voice

($N=1$). The patients on amitriptyline ($N=1$), escitalopram ($N=1$), and sertraline ($N=1$) perceived a significant effect with no side effects and are still treated with these antidepressants. See Table 5 for the CGI-C scores of the antidepressants. One patient treated with citalopram perceived improvement in mood but experienced also side effects and treatment was stopped. In 33.3% of patients treated with fluoxetine the condition was much improved (score 2 on CGI-C), in 72.2% the treatment had at least minimal effect (score 2 or 3 on CGI-C), in 16.7% there was no effect (score 4 on CGI-C), and in one patient (5.6%) the condition was much worse (score 6 on CGI-C).

3.4.3. Antipsychotics

Twelve patients received treatment with antipsychotics (14 unique treatments were started). One patient was treated first with pipamperone but switched to risperidone. Another patient started with risperidone and changed to aripiprazole. Average age of onset of neurobehavioral symptoms was 9.0 years ($SD= 4.9$) and average age of start PD treatment was 12.1 years ($SD= 5.2$) in this group.

Patients treated with antipsychotics had the highest number of symptoms (mean= 5.1, $SD= 1.9$, range= 2–8), with emotional

and behavioral dysregulation ($N=9$) and autistic features ($N=9$) reported the most.

Half of the patients in our sample treated with antipsychotics are currently still treated with the same PD. Treatment was terminated in six patients due to no perceived effect (42.9%). In one patient treatment with risperidone was successful and after three years the treatment could be stopped gradually without return of neurobehavioral symptoms. One of our patients reported more anxiety after increasing the dose of risperidone. Table 5 summarizes the CGI-C scores for antipsychotics. Risperidone had no effect in 22.2% of the patients (score 4 on CGI-C), minimal effect in at least 66.7% (score 2 or 3 on CGI-C), and in 22.2% of the patients the condition was much improved (score 2 on CGI-C). One patient's condition was minimally worse after treatment with pipamperone (score 5 on CGI-C), another patient's condition was minimally to much improved (score 2–3 on CGI-C), and in one patient there was almost no effect (score 4 on CGI-C). Aripiprazole treatment very much improved (score 1 on CGI-C) the condition of one patient and had no effect on another's condition (score 4 on CGI-C).

3.4.4. Tranquilizers

Two patients were treated with tranquilizers (alprazolam and oxazepam). These patients were somewhat older, namely 21.5 years old ($SD= 2.1$). One patient had two symptoms (CGI-S = 6, severely ill) and the other seven symptoms (CGI-S = 4.5, between moderately and markedly ill). Both patients are still using tranquilizers. No side effects were reported and no changes in dosing were made. Table 5 shows the CGI-C scores. Alprazolam

treatment much improved the patient's condition (score 2 on CGI-C), oxazepam had minimal effect (score 3 on CGI-C).

4. Discussion

The aim of this study is to describe the clinical experience of psychopharmaceutical interventions to alleviate neurobehavioral symptoms in a sample of 52 DMD patients.

4.1. Indications for PD-treatment

Most PD-treated DMD patients suffered from four or more neurobehavioral symptoms and one fifth had even six neurobehavioral symptoms or more. This is far more than reported in a similar study by Darmahkasih A.J., Rybalsky I., Tian C., Shellenbarger K.C., Horn P.S., Lambert J.T. et al. in which 22.6% of all DMD patients (N=700) had four or more neurobehavioral symptoms [13]. This suggests that especially the more complex neurobehavioral profiles are considered for PD-treatment. The neurobehavioral profiles seen in this study are not representative for the whole DMD population, but seem similar to findings in earlier research, with inattentive and hyperactive features and emotional and behavioral dysregulation as being the most frequently reported neurobehavioral symptoms in DMD [20,21]. Despite similar severity scores, behavioral profiles in our sample seem different in DMD patients treated with prednisone than in DMD patients treated with deflazacort. Anekdotical evidence suggest different behavioral side effects of both corticosteroid treatments, but further research is needed to address this more in depth [22,23]. Global intellectual delay was less frequently observed in our PD-treated population (11.5%) than we would expect based on findings in the general DMD population, in which prevalence of intellectual disabilities is rated in the range of 19–26% [21]. This could suggest that PDs are more frequently prescribed for DMD patients without global intellectual delay. The reason for this disparity should be subject of future research. Interestingly, sleep problems were very frequently reported in this cohort with severe neurobehavioral symptoms (51.9%), whereas studies reporting about sleep problems are rare.

4.2. Psychopharmaceutical interventions

DMD patients with inattentiveness, hyperactivity features, learning problems and emotional and behavioral dysregulation as predominant symptoms, were mostly treated with methylphenidate, the first-choice PD-treatment for this kind of neuropsychiatric difficulties [24]. Methylphenidate has been proven to be effective in numerous studies and meta-analyses in many different populations [25–27], but treatment can also be compromised by a range of adverse effects like elevated blood pressure and heart rate, restlessness, sleep difficulties, reduced appetite and weight loss, headaches and stomach pain [28]. Especially in the DMD population thorough medical screening prior to the initiation of PD treatment is essential to exclude contraindications and detect already existing symptoms like gastrointestinal issues, weight problems, chronic pain, and fatigue [29–31]. In contrast with the bulk of evidence in other populations, only two studies addressed methylphenidate treatment in DMD patients [12,13]. The retrospective patient record review of Darmahkasih et al. (N=86/700) only minimally describes methylphenidate treatment, whereas Lionarons et al. do report about a systematic follow-up after short-acting methylphenidate treatment in DMD (N=10) [12,13]. They found methylphenidate to be effective in their cohort, with only minimal side effects [12]. These findings are in line with our observational results. Nonetheless, side effects and rebound effects when

methylphenidate wears off can occur and good follow-up and dosage management on an individual level seem necessary. It may be important to monitor growth evolution, systolic blood pressure and cardiac functioning closely, as a recent review concluded that long-term treatment with methylphenidate may be associated with a slight growth deficit, an elevated systolic blood pressure and an increased heart rate [32,33]. In our cohort, however, no signs of cardiac side effects were observed. Altogether, treatment with methylphenidate seems to be a possible solution for DMD boys with inattentiveness, hyperactive features, learning problems and emotional and behavioral dysregulation when good management of side effects and rebound effects, as well as therapeutic goals are achieved.

DMD patients with obsessive compulsive features, depressive symptoms, and/or anxiety problems were most often treated with fluoxetine (81%), which is in line with international guidelines [34]. A third of the patients treated with SSRIs showed much improvement and over 40% showed minimal improvement. It is possible that the daily dose is lower than the recommended range of 20–60 mg daily in a pediatric population, and may have led to differences in therapeutic responses [35]. As higher doses are not always associated with more significant responses, further research should look into dose management or other therapeutic possibilities to effectively treat these kind of symptoms within the DMD population [36].

Antipsychotics were mostly prescribed in patients with autistic features combined with emotional and behavioral issues and showed varying results. No side effects were reported during risperidone treatment, but the perceived success in our cohort was variable with a third showing much improvement, a third showing minimal improvement and a third showing no effect. A good medical follow-up during risperidone treatment is recommended as weight gain and loss of appetite may occur [37].

One patient was successfully treated with apripazole which is in line with the case report of Noda et al. (2021), however treatment in another patient showed no effect [38].

In general, our results show that younger boys (around eight years of age) with DMD are more often treated with methylphenidate and show more hyperactive and inattentive behavior. As they grow older, boys with DMD are more often treated with fluoxetine and an emotional behavioral and obsessive compulsive profile is observed more often. They may grow out of attention deficits or it could be possible that with the loss of physical ability other neurobehavioral symptoms come to surface. It would be of high value to investigate if the functional status, loss of agency and/or illness perceptions are associated with a higher prevalence of certain behavioral symptoms. However, as far as we know, this phenomenon has not been explored before and needs to be further evaluated in prospective studies.

4.3. Limitations

We believe that the present findings can be of help for clinicians working with DMD patients, when psychiatric symptoms interfering with development appear, but methodological limitations of our study must be taken in mind. This was a retrospective, descriptive study and therefore general conclusions or recommendations for the whole DMD population cannot be made. Additionally, more in depth analysis of subgroups was not possible. This would be interesting for future research. Another limitation is the validity of retrospective use of the CGI scales, which, as argued by Forkmann et al. (2011), implies that clinicians must remember a patient's condition before starting medication and then contrast it to their current state [39]. The results may therefore be distorted by the psychological phenomena called 'hindsight bias' [40]. This may have led to rating patients as more

impaired at admission than they actually were. Related to this, we found minimal agreement between the raters on both CGI-S and CGI-C scales. A possible explanation for this is the retrospective nature of this study together with the fact that the items of the CGI are open for interpretation and responses on the CGI represent a clinical judgment rather than an absolute score [19]. Despite a shared expertise in DMD, the specific specialization (psychiatry/psychology versus neurology) could have influenced the impression of the different professionals. This discrepancy in clinical judgement underscores the importance of more systematic and standardized care for mental health problems in DMD. Our suggestion would be to closely monitor favorable and adverse effects when PD-treatment is considered. More insight is needed on how to measure these outcomes in a clinical setting.

Finally, our study design does not allow us to draw conclusions about psychopharmaceutical co-therapy in DMD.

5. Conclusion

The aim of this study was to describe the clinical experience of psychopharmaceutical interventions in a sample of 52 DMD patients. Our results accentuate that the DMD population is a heterogeneous population with a complex overlap in neurobehavioral symptoms, which can evolve during developmental stages and disease progression. Treatment with psychopharmaceuticals may improve most of these neurobehavioral symptoms, but the success of the treatment seems to be case dependent and side effects need to be monitored carefully and multidisciplinary. In spite of the retrospective descriptive design, our study highlights the need for more systematic and standardized neurobehavioral assessment and support for DMD patients and their families.

Declaration of Competing Interest

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

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